2026 Workshop: Updating urban segregation computation with R

9-13th November 2026

How - and why - to model segregation as a temporal (everyday) phenomenon ?

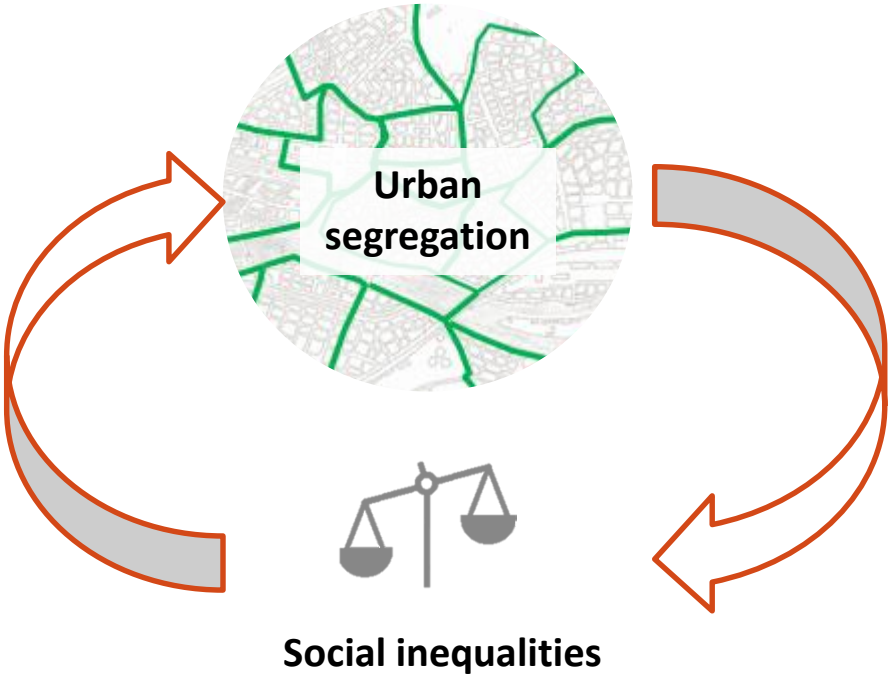
Julie Vallée

CNRS, Toulouse, France

ABOUT SEGREGATION

A **spatial expression**
of social inequalities

A **spatial driver**
of social inequalities



strategies of the dominant social class for controlling geographical space

neighbourhood effects

RESIDENTIAL SEGREGATION



- Residential segregation
  - Based on people' places of residence
  - 1 person = 1 place of residence

- Residence
  - a pivot of individuals' daily lives.
    - ⟶ the place for physiological activities (sleep, meals, etc.)
    - ⟶ the main location for family relationships
  - But a location unequally occupied during the day depending on social groups

➔ *An important place, but not the only one…*

WHY TO MODEL … ?

EVERYDAY MOBILITY



PEOPLE'S DAILY MOBILITY



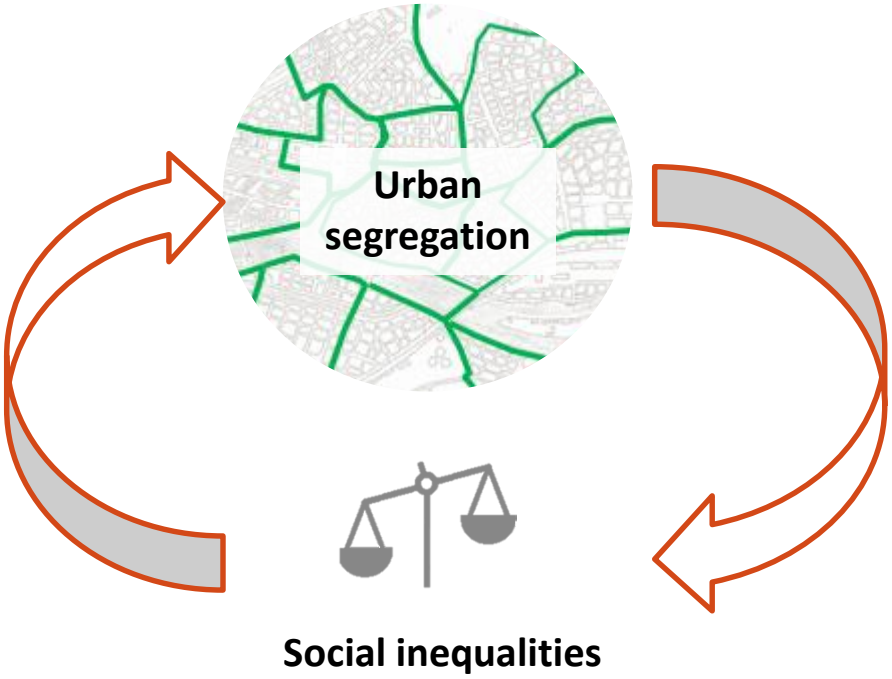


DAILY VARIATIONS IN NEIGHBORHOOD'S SOCIAL COMPOSITION AND IN SEGREGATION

EVERYDAY SEGREGATION

A **spatial and temporal expression** of social inequalities

A **everyday driver** of social inequalities



*strategies of the dominant social class for controlling **time** & space*

*neighbourhood effects **around the clock***

IN URBAN STUDIES…

A surprising discrepancy between:

- the high interest in measuring people's daily mobility (from surveys, GPS sensors or mobile phones)

and

- the low interest towards hourly variations in social composition of neighbourhoods
  - to measure neighbourhood effects by linking hourly neighbourhood features with people's timetables
  - to shed new light on urban segregation

"*About the importance of the 'daycourse of places' and the daycourse of place effects to gain insight into neighbourhood effects and their hourly dynamics*"


The daycourse of place

Julie Vallée

Social Science & Medicine 194 (2017) 177–181

https://doi.org/10.1016/j.socscimed.2017.09.033

HOW TO MODEL … ?

EVERYDAY LOCATIONS: ACTIVITY *VS.* HOURLY LOCATIONS

Locating people in geographical space, according to…

- … their **activity locations**

  eg, their workplaces. ⟶ *"Occupational segregation"*

  - Partial view of the population (only active individuals) and their mobility
  - 1 person = 0 location (inactive); 1 location (single-site active); 2 or more locations (multi-site active…)

- … their **spatial location at a given time**
  - 1 person = 1 location (no more, no less)
  - requires space-time data (≠ census)
  - Cf. *time-geography* approach

HOW TO MODEL … ?

SUMMARY OF DATA SOURCES FOR EXPLORING HOURLY VARIATIONS IN URBAN SEGREGATION

	Census	**Mobile phone**	**Mobile phone apps with GPS**	**Ticketing**	**Travel survey / OD survey**
Mode of data collection	*Self-reporting*	Passive	Passive	Passive	Self-reporting
Collection frequency	*one-time*	continuous	continuous	continuous	one-time
Locations	*Residential and work places*	All locations (phone on)	All locations (Apps and GPS in active mode)	Origin (and destination) of each trip (in the transport system)	Origin and destination of each trip (on the day of the survey)
Spatial accuracy	*Census tracts*	Cell tower	X; Y	X; Y (stations)	**Districts**
Temporal accuracy	***No***	hh:mn	hh:mn	hh:mn	hh:mn
Personal information	*Yes*	**No (or very few)**	**No (or very few)**	**No (or very few)**	Yes
Trip information (mode & purpose)	***No***	**No**	**No**	Mode : Yes Purpose : **No**	Yes
Sample size	*Quasi-exhaustive*	Often very large	It depends (but no representative)	Often very large	Small (but representative)

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HOW TO MODEL … ?

ABOUT MOBILE PHONE DATA

- Respondent's sample
  - Really big?

    Not necessarily since only complete data can be used (e.g. respondents for which residential areas can be located)
  - Socially representative?

    No, and even less once only respondents with complete data are considered
  - Socially informative?

    No… and be carefull about perillous social inferences/imputations based on residential areas

" A notable limitation of this study is a lack of socioeconomic information about the individual phone users. Socioeconomic data was joined ecologically with the underlying assumption that a phone represents the socioeconomic features of the neighbourhood where it was during night-time (assumed night rest location). This introduces a degree of imprecision in our estimates while protecting the privacy of phone users "

Toger M, Türk U, Östh J, Kourtit K, Nijkamp P. 2023. Inequality in leisure mobility: An analysis of activity space segregation spectra in the Stockholm conurbation. *Journal of Transport Geography*

HOW TO MODEL … ?

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- Geographical attributes
  - Spatial granularity?

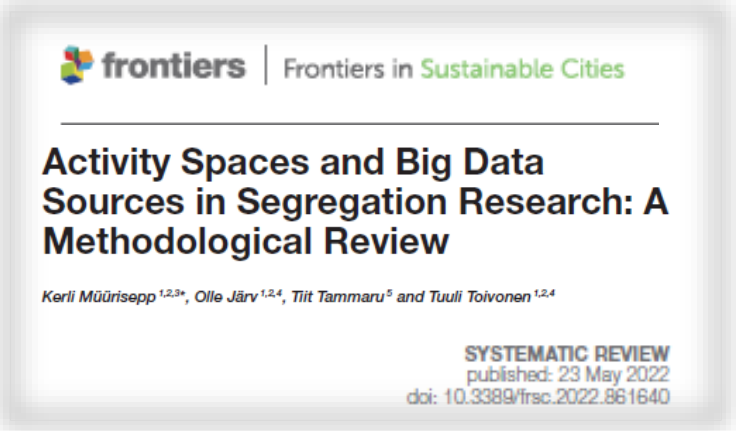
  According to antenna coverage (i.e. to people's density)

- Temporal attributes
  - Frequency?

  Often centered on one short period
  - Recent ?

  It depends

See also :



HOW TO MODEL … ?

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?

LIMITS AND STRENGTHS

	Travel survey / OD survey
Mode of data collection	Self-reporting
Data-collection frequency	one-time
Locations	Origin and destination of each trip (on the day of the survey)
Spatial accuracy	Districts
Temporal accuracy	hh:mn
Personal information	Yes
Trip information (mode & purpose)	Yes
Sample size	Small (but representative)

Limitations related to…

>> the **sample size:**

- Risky to cross **multiple socio-demographic indicators**
- Risky to drill down to a **spatial granularity finer than the district**
- Only presence of **residents** in the region

>> the **frequency** of data collection:

- Impossible to study **weekly, monthly, or annual variations**

And strengths:

- Information on **purposes, modes of transport**, and **individual socio-demographic profiles** without risky interpolations
- Good **representativeness of the population** in the studied territories
- **Reproducibility & international comparison**

HOW TO MODEL … ?

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References

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